Deep reinforcement learning

Unit 6 · reference sheet · AI/ML in 35 topics

Why tables die

$|S| \times |A|$: grid 25, chess $\sim 10^{44}$, Atari 256^{100800}

Deadly triad

function approximation + bootstrapping + off-policy $\Rightarrow$ divergence
Any two are safe. All three is where DQN lives.

DQN

Replay: sample random minibatches $\Rightarrow$ breaks temporal correlation.
Target net: $\theta^- \leftarrow \theta$ every $\sim 10^4$ steps $\Rightarrow$ fixed target.
$L = \left(r + \gamma \max_{a'} Q_{\theta^-}(s', a') - Q_\theta(s, a)\right)^2$ (Huber)
Double DQN: select with θ, evaluate with θ^-.

Policy gradient

$\nabla J = E[\nabla_\theta \log \pi_\theta(a \mid s)\, A(s, a)]$
Baseline $b(s)$: unbiased, lower variance; $A = Q - V$
PPO: $\min(rA,\ \text{clip}(r, 1-\varepsilon, 1+\varepsilon)A)$, $\varepsilon = 0.2$
$r = \pi_{new}/\pi_{old}$; clip bounds the objective, not the weights.

Gotchas

- Replay + on-policy method = invalid gradient.
- PPO is sample-inefficient by design (on-policy).
- Overestimation comes from max over noisy Q; Double reduces, not removes.

This unit at a glance

#	Topic	You can
23	Function approximation	explain why tables die and why approximation kills convergence guarantees
24	DQN	explain replay buffer + target network as fixes for correlated data and moving
25	Policy gradient at scale	trace REINFORCE → baseline → A2C/PPO and explain the variance problem

If you see this → do that

Signal	Move
Variance too high	baseline / GAE
Updates destroy policy	clip (PPO)
Continuous control	PPO / SAC
Sparse reward	shaping / curriculum
Need replay	off-policy only

Numbers worth memorising

Replay $\sim 10^6$; target sync $\sim 10^4$ steps; batch 32
PPO $\varepsilon = 0.2$; ratio 1.3 → clipped 1.2
Advantage $A = Q - V$; GAE $\lambda \in [0.9, 0.95]$
DQN frame stack 4 (velocity is not in one frame)